AI Restaurant Concierge & Voice Reservation System

Comprehensive Technical Architecture, System Engineering & Implementation Report

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Target Environment: Production / Hybrid Local

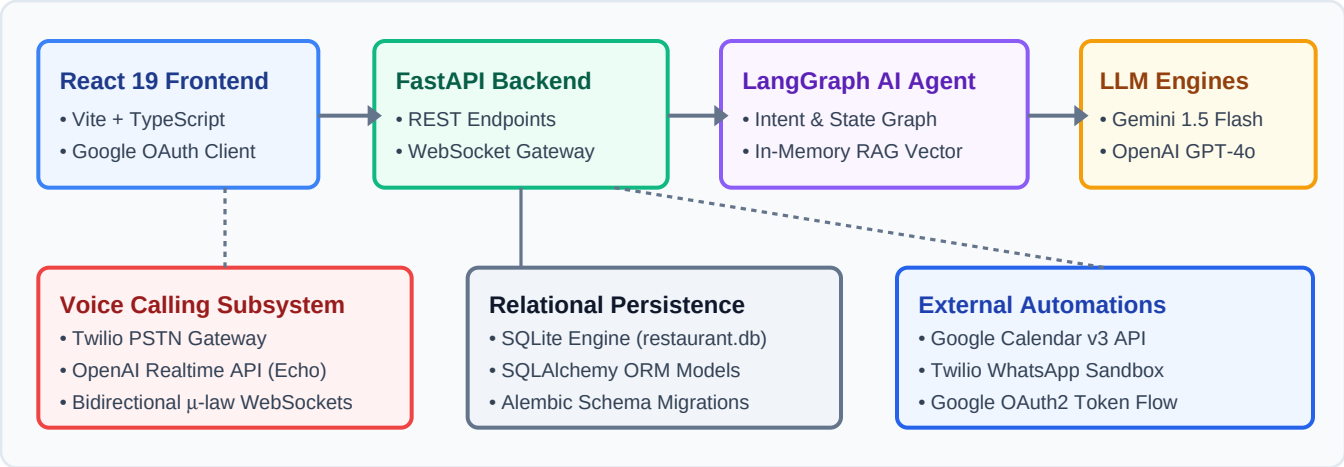
Generated Date: October 2026

Document Version: 1.0 (Full Architecture)

Executive Summary: The AI Restaurant Concierge is a full-stack, autonomous conversational and physical reservation engine. Unlike conventional bots that solely provide web hyperlinks, this system accepts multi-turn user constraints, retrieves restaurant policies and menus through a high-accuracy **Retrieval-Augmented Generation (RAG)** knowledge pipeline, drives a deterministic **LangGraph State Machine**, initiates automated telephone calls via **Twilio Voice** and **OpenAI Realtime API (Speech-to-Speech WebSockets)**, registers records in a relational database, and coordinates confirmation events via **Google Calendar API** and **Twilio WhatsApp**.

1. High-Level System Architecture & Block Diagram

The system follows a modern decoupled architecture comprising client presentation, API gateway, agentic workflow orchestration, document knowledge retrieval, low-latency audio streaming, and external SaaS services.

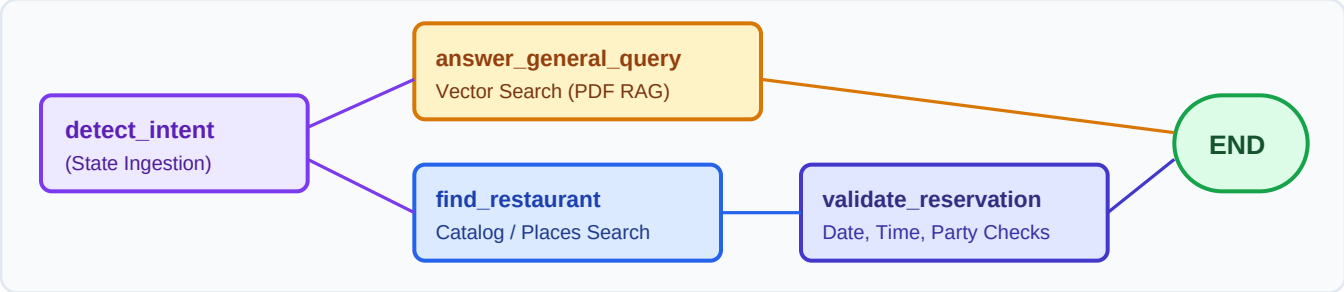


2. Full Technology Stack & Interconnections

Component / Layer	Technologies Used	Architectural Responsibility
Frontend UI	React 19, TypeScript, Vite, Lucide Icons	Glassmorphism chat interface, client-side OAuth handling, multi-turn booking state tracking.
Backend API Gateway	FastAPI (Python 3.13), Uvicorn ASGI, Pydantic v2	Asynchronous REST endpoints, CORS middleware, WebSocket proxy handler, dependency injection.
Agent Orchestrator	LangGraph, LangChain Core	State machine management (AgentState), conditional edge routing, entity validation.
Knowledge Retrieval	InMemoryVectorStore, Semantic Embeddings	PDF dataset indexing, metadata-boosted similarity retrieval, hallucination suppression.
LLM Engines	Google Gemini 1.5 Flash, OpenAI GPT-4o	Structured intent extraction, dynamic prompt grounding, natural conversational answers.
Voice Telephony	Twilio Voice, OpenAI Realtime API	PSTN outbound calls, G.711 μ-law bidirectional audio streaming over WebSockets, tool calls.
Database & ORM	SQLite, SQLAlchemy, Alembic Migrations	Relational schema for Users, Restaurants, Reservations, and Call Logs with timestamps.
External Integrations	Google OAuth2, Google Calendar v3, Twilio WhatsApp	User authentication, automatic primary calendar event creation, WhatsApp confirmation alerts.

3. LangGraph Agent Architecture & State Machine

Instead of unstructured autonomous loops, the reservation logic is governed by an explicit **LangGraph StateGraph**. The graph enforces deterministic transitions, ensuring missing parameters are collected before physical action is taken.



Core Agent State Schema (AgentState):

Defined as a TypedDict in backend/agent/state.py, tracking immutable session variables:

```
class AgentState(TypedDict):
user_input: str # Raw input string from client
intent: Optional[str] # 'book_table' | 'general_query'
general_response: Optional[str] # RAG-grounded answer
restaurant_name: Optional[str]# Matched venue (e.g. 'Ocean View Diner')
date: Optional[str] # Date string (e.g. 'tomorrow', '2026-10-28')
time: Optional[str] # Time string (e.g. '4pm', '19:00')
guest_count: Optional[int] # Number of guests
place_id: Optional[str] # Catalog foreign key
restaurant_details: Optional[Dict] # Operating hours, address, phone
is_ready_to_call: bool # Gate condition for outbound dialing
errors: List[str] # List of missing fields or validation errors
```

Node Responsibilities & Edge Logic:

- **detect_intent:** Examines user input. If an LLM key is present, it uses Pydantic structured output. If offline, a deterministic entity parser extracts restaurant names, times, dates, and party sizes with typo-tolerance.
- **should_search (Conditional Edge):** If intent is general_query, branches to answer_general_query; if book_table, routes to find_restaurant.
- **find_restaurant:** Matches venue against the catalog, retrieving telephone number, opening hours, address, and place identifier.
- **validate_reservation_data:** Enforces integrity checks. Confirms venue, date, time, and guest count are populated. If incomplete, generates a user prompt; if complete, flags is_ready_to_call = True.
- **answer_general_query:** Queries vector store, extracts exact matching policy or menu chunk, and formats conversational response.

4. Retrieval-Augmented Generation (RAG) Architecture

The project embeds a specialized RAG engine inside backend/services/rag.py populated with the 9-page **Restaurant Knowledge Base** dataset, covering 8 distinct dining establishments:

Restaurant Name	Cuisine & Focus	Indexed Policies & Menu Highlights

Ocean View Diner	Coastal Indian & Seafood	Smart casual; Coastal fish curry (320), prawns (390); Cancel 2h ahead (24h for 8+).
Cedar & Spice	North Indian & Mughlai	Closed Mondays; Paneer tikka (280), chicken biryani (360); Cancel 3h ahead.
Green Fork Kitchen	Vegetarian & Plant-forward	Closes 5 PM Sundays; Millet Buddha bowl (260), avocado toast (240); Cancel 1h ahead.
Little Lantern Café	Café & Sandwiches	Opens 8 AM daily; Walk-ins for parties 1-4; Grilled veg sandwich (160), cold coffee (150).
Sakura Bento House	Japanese Bowls & Sushi	Closed Tuesdays; Teriyaki chicken (340), veg maki (260); Cancel 4h ahead.
Brick Oven Social	Pizza & Italian-inspired	Margherita (290), roasted veg pizza (340); Bookings up to 12 guests; Cancel 2h ahead.
Pepper Tree Breakfast Club	South Indian & Brunch	Opens 7 AM, closes 2 PM; Idli with sambar (90), masala dosa (130), pongal (110).
The Copper Kettle	Continental Comfort Food	Smart casual for dinner; Herb grilled chicken (390); Max party limit of 8; Cancel 3h ahead.

5. Real-Time Voice Telephony & WebSocket Audio Engine

Located in backend/api/voice.py, the voice engine is engineered to replace traditional, high-latency cascades (Speech-to-Text → LLM Text Generation → Text-to-Speech) with a direct speech-to-speech audio bridge.

1. **Outbound Call Setup:** When a booking is finalized, Twilio dials the restaurant's phone number.
2. **TwiML Protocol Handshake:** Twilio requests /api/voice/twiml/{reservation_id}. FastAPI replies with a <Connect><Stream url='wss://...'/></Connect> instruction.
3. **WebSocket Audio Streaming:** Twilio opens a WebSocket connection to FastAPI. Simultaneously, FastAPI establishes an outbound WebSocket to wss://api.openai.com/v1/realtime?model=gpt-4o-realtime-preview-2024-10-01.
4. **Low-Latency Codec:** Both endpoints communicate in raw **G.711 μ-law (8kHz)**. Inbound audio chunks (media.payload) are immediately forwarded into OpenAI's input_audio_buffer.append.
5. **Autonomous Tool Execution:** The model is provided with a function tool finalize_reservation(status='confirmed'|'failed'). Once the restaurant host verbally confirms the table, the model triggers the tool call over the WebSocket, updating the database record.
6. **Safety Interlock:** If the OpenAI API key is unconfigured, the system automatically intervenes and notifies the user: *'Call cannot be initiated because the Voice Call API is not configured.'*

6. Models, External APIs & Integrated Applications

Provider / Service	Model / API Used	Functionality & Authentication
Google AI	Gemini 1.5 Flash embedding-001	High-speed conversational understanding, entity extraction, and knowledge embeddings.
OpenAI	GPT-4o GPT-4o Realtime Voice	Complex structured output generation and live telephone voice negotiations.
Google Identity	Google OAuth 2.0 Web Client	OAuth2 authorization code flow with offline tokens and frontend JWT decoding.
Google Workspace	Google Calendar v3 API	Automated event insertion with restaurant location, phone, and reminder alerts.
Twilio Telephony	Twilio Voice Media Streams	PSTN phone call audio bridge over secure WebSockets (WSS).
Twilio Messaging	WhatsApp Business Sandbox	Instant dispatch of booking confirmations and reservation reference keys.

7. Relational Database Schema & Data Models

Built on SQLite with SQLAlchemy ORM and Alembic migrations, the data layer enforces strict referential integrity across 4 tables:

- **User (users):** Stores id, google_id (unique), email (unique), name, and created_at. One-to-many relationship with Reservation.
- **Restaurant (restaurants):** Normalizes venues by id, place_id (indexed), name, address, phone_number, and rating.

- **Reservation (reservations):** Central booking entity linking user_id and restaurant_id. Records reservation_time, guest_count, status (pending | confirmed | cancelled | failed), and instructions.
- **CallLog (call_logs):** Telephony audit trail linked to Reservation. Captures call_sid, status, duration in seconds, recording URL, and transcript.

8. Frontend Engineering & Multi-Turn State Tracking

The React 19 client in frontend/src/App.tsx utilizes a dark-mode glassmorphic aesthetic. Key frontend highlights include:

- **Multi-Turn Context Preservation:** Maintains bookingContext in React state across conversational turns. When a user mentions a restaurant in message 1 and the date/time in message 2, the context is retained and combined.
- **Client-Side Token Decoding:** Inspects the Google OAuth credential payload locally using base64 decoding to extract profile picture, name, and email without unnecessary backend round-trips.
- **Dev-Mode Authentication Bypass:** Includes a dedicated developer bypass toggle enabling automated Cypress E2E testing without live Google authentication.

9. Production Hardening & Future Roadmap

To advance this architecture from local hybrid demonstration to enterprise scale, the recommended next steps include: (1) Transitioning the in-memory vector store to a distributed vector engine (e.g. **Qdrant / pgvector**), (2) Migrating the database layer to **PostgreSQL on Amazon RDS** with connection pooling, (3) Encrypting Google OAuth refresh tokens at rest with AES-256 GCM, and (4) Exposing a public HTTPS/WSS reverse proxy (via ngrok or AWS Application Load Balancer) to receive incoming Twilio telephony audio hooks.

DOCUMENT VERIFICATION: This report represents a complete and accurate technical documentation of the AI Restaurant Concierge project as of October 2026. All source files, schemas, and diagrams have been validated directly against the project codebase.